Requirements Document

Faculty Consultation Booking System

Midterm Project — Campus System, Round 1

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GROUP 6

Project Overview

The Faculty Consultation Booking System is a web application that lets students request and schedule consultation time with faculty members, and lets faculty publish their available consultation slots and manage incoming requests. The system replaces informal, ad hoc scheduling (walk-ins, messaging apps, posted office hours) with a single, role-based platform that tracks availability and booking status end to end.

The problem being addressed is that students currently have no reliable way to see when a faculty member is actually available for consultation, and faculty have no structured way to receive, approve, or track consultation requests. This leads to missed meetings, double-booked time, and no record of whether a request was ever addressed.

Users

People who directly interact with the system through its interface.

Role	Description
Student	Browses faculty-posted consultation slots, books a slot by stating a purpose, and tracks the status of their own bookings (pending, approved, declined, done, cancelled).
Faculty	Posts open consultation slots (date, start time, end time), and reviews, approves, declines, or completes booking requests submitted by students.
Admin	Oversees the system: views all registered users and all bookings across every student and faculty member, for monitoring and support purposes.

Stakeholders

People or groups who are affected by, or have an interest in, the system, even if they do not use it directly.

Stakeholder	Relationship to the System
Students	Primary users; need a reliable, transparent way to reach faculty for consultation.
Faculty members	Primary users; need an organized way to manage their consultation availability and requests without being overwhelmed by informal messages.
Department Chair / Registrar's Office	May rely on consultation records as evidence that academic support is being provided to students.
Course Instructor (Mr. Mamansag)	Evaluates the system as a midterm project deliverable and assesses its functionality, requirements fit, and development process.
Development Team	Responsible for building, testing, maintaining, and defending the system.

Functional Requirements

What the system must do.

ID	Requirement
FR-1	The system shall allow a new user to register an account as either a Student or Faculty, with a hashed password stored in the database.
FR-2	The system shall allow a registered user to log in with their email and password, establishing a session.
FR-3	The system shall allow a logged-in user to log out, ending their session.
FR-4	The system shall allow a Faculty user to create a consultation slot specifying date, start time, and end time.
FR-5	The system shall allow a Student user to view a list of all currently available (unbooked) consultation slots, including the faculty member's name and department.
FR-6	The system shall allow a Student user to book an available slot by submitting a stated purpose for the consultation.
FR-7	The system shall prevent a slot from being booked more than once; once booked, it shall no longer appear in the list of available slots.
FR-8	The system shall allow a Faculty user to view all bookings made against their own slots.
FR-8	The system shall allow a Faculty user to approve, decline, mark as

	done, or cancel a booking request.
FR-10	FR-10 The system shall allow a Student user to view the status of all of their own bookings (pending, approved, declined, done, cancelled).
FR-11	The system shall allow an Admin user to view a list of all registered users in the system.
FR-12	The system shall allow an Admin user to view a list of all bookings made across the entire system.
FR-13	The system shall restrict every protected page so that it is accessible only to users with the appropriate role (Student, Faculty, or Admin), and shall redirect or deny access to an unauthorized user who attempts to access it, including by typing the URL directly.

Non-Functional Requirements

How the system should perform.

ID	Requirement
NFR-1	Security: User passwords shall be hashed with bcrypt before being stored; plain-text passwords shall never be stored or logged. Session data shall be used to track authentication state rather than exposing credentials in the URL.
NFR-2	Data Integrity: All data submitted through forms shall be validated (required fields, duplicate email detection) before being stored, and shall be stored using parameterized SQL queries to prevent SQL injection.
NFR-3	Persistence: All submitted data (accounts, slots, bookings) shall be stored in a MySQL database such that it persists across server restarts and page refreshes.
NFR-4	Usability: The interface shall use plain, consistent navigation so that a first-time user can register, log in, and complete their primary task (posting or booking a slot) without external instructions.
NFR-5	Availability: The application shall run reliably on a local XAMPP/Node.js environment for the duration of a live demonstration without requiring a restart.
NFR-6	Maintainability: The codebase shall follow an MVC-style structure (routes, controllers, models, views) so that individual features can be modified independently.

Use Cases

Each use case describes a major system function in terms of the actor involved, the precondition that must be true before it can occur, the steps taken, and the expected result.

UC-1: Register an Account

Field	Description
Actor	Student or Faculty (unregistered visitor)
Precondition	The visitor does not yet have an account, and is on the home page.
Steps	1. The visitor clicks "Register" from the home page. 2. The visitor enters their full name, email, password, selects a role (Student or Faculty), and optionally a department. 3. The visitor submits the form. 4. The system checks whether the email is already registered. 5. The system hashes the password and stores the new account.
Expected Result	A new user account is created, and the visitor is redirected to the login page. If the email is already registered, an error message is shown and no account is created.

UC-2: Log In

Field	Description
Actor	Registered user (Student, Faculty, or Admin)
Precondition	The user has a registered account and is on the login page.
Steps	1. The user enters their email and password. 2. The user submits the form. 3. The system looks up the account by email and compares the submitted password against the stored hash. 4. If the credentials are valid, the system creates a session storing the user's ID, name, and role.
Expected Result	The user is redirected to their role-appropriate dashboard. If the credentials are invalid, an error message is shown and the user remains on the login page.

UC-3: Post a Consultation Slot

Field	Description
Actor	Faculty
Precondition	The Faculty user is logged in.
Steps	1. The Faculty user navigates to "Post a New Consultation Slot" from their dashboard. 2. The Faculty user enters a date, start time, and end time. 3. The Faculty user submits the form. 4. The system stores the new slot, linked to the Faculty user's account, with its booked status set to false.
Expected Result	The new slot becomes visible to students on the available slots page. A

	Student user attempting to reach this page without a Faculty role is denied access.

UC-4: Book a Consultation Slot

Field	Description
Actor	Student
Precondition	The Student user is logged in, and at least one unbooked slot exists.
Steps	1. The Student user navigates to "Browse Available Slots." 2. The Student user selects a slot and enters a purpose for the consultation. 3. The Student user submits the booking form. 4. The system re-checks that the slot is still unbooked, then creates a booking record with status "pending" and marks the slot as booked.
Expected Result	The booking appears in the student's "My Bookings" list with status "pending," and the slot no longer appears on the available slots page. If the slot was booked by someone else in the meantime, the system informs the student and does not create a duplicate booking.

UC-5: Manage a Booking Request

Field	Description
Actor	Faculty
Precondition	The Faculty user is logged in and has at least one incoming booking.
Steps	1. The Faculty user navigates to "Manage Incoming Bookings." 2. The Faculty user reviews a pending booking's purpose and requested time. 3. The Faculty user clicks Approve or Decline. 4. For an approved booking, the Faculty user may later click Mark as Done or Cancel.
Expected Result	The booking's status is updated accordingly, and the new status is immediately visible to the student on their "My Bookings" page.

UC-6: View System-Wide Records

Field	Description
Actor	Admin
Precondition	The Admin user is logged in.
Steps	1. The Admin user navigates to "Manage Users" or "View All Bookings" from their dashboard. 2. The system queries all user or booking records from the database.
Expected Result	The Admin user sees a complete table of every registered user, or every booking in the system, regardless of which student or faculty account

	created it. A Student or Faculty user attempting to reach these pages is denied access.

Access Control Summary

Role-based access is enforced through server-side middleware rather than by hiding links in the interface, so a user cannot bypass it by typing a URL directly.

Page	Student	Faculty	Admin
Post a consultation slot	Denied	Allowed	Denied
Browse / book available slots	Allowed	Denied	Denied
Manage incoming bookings	Denied	Allowed	Denied
View all users / all bookings	Denied	Denied	Allowed
Dashboard (own view)	Allowed	Allowed	Allowed